



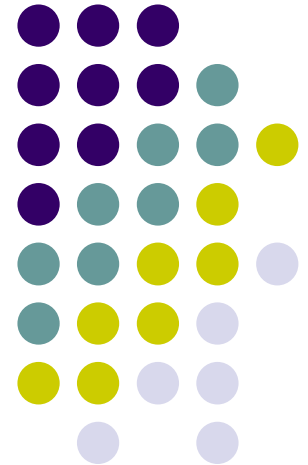
ຄະນະວິທະຍາສາດທຳມະຊາດ
ພາກວິຊາ ວິທະຍາສາດຄອມພິວເຕີ

ເວບເຊີເວີ ແລະ ເຕັກໂນໂລຢີ (Web Server and Technology)

ສອນໂດຍ:

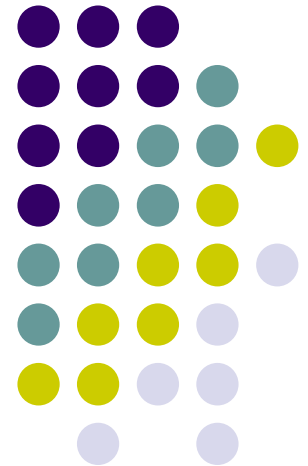
ມີຖື:

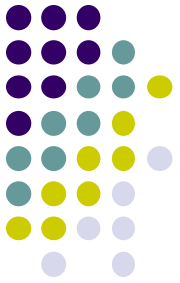
ອີເມວ:



ບົດທີ10

ເຕັກໂນໂລຊີຖານຂໍ້ມູນ (Database Technology)





ເນື້ອໃນໂດຍລວມ

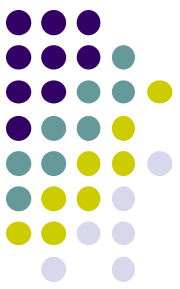
- ຖານຂໍ້ມູນ MySQL
- ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL
- ການຈັດການຖານຂໍ້ມູນ
- ການນຳໃຊ້ PHP ເຊື່ອມຕໍ່ຖານຂໍ້ມູນ
- ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows

ຖານຂໍ້ມູນ MySQL



- MySQL ເປັນໂປຣແກຣມຈັດການລະບົບຖານຂໍ້ມູນເຊິ່ງສໍາພັນ (Relational Database Management System) ຊຶ່ງສະໜັບສະໜູນການໃຊ້ວຽກໃນລະບົບປະຕິບັດການ ເຊັ່ນວ່າ: Unix, Mac ແລະ Window ນອກນີ້ຍັງເຮັດວຽກຮ່ວມກັບ Java, C, C++, PHP, ASP, Perl ແລະ ອື່ນໆ
- MySQL ໄດ້ຮັບການພັດທະນາຕາມແນວທາງມາດຕະຖານຂອງພາສາ SQL, ສະນັ້ນສາມາດນໍາໃຊ້ພາສາ SQL ໃນການໃຊ້ວຽກໄດ້.

ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL



- ການຕິດຕັ້ງ MySQL

```
#yum install mysql mysql-server ຫຼື mariadb-server
```

- ຄໍາສັ່ງຈັດການ MySQL

```
#service mysqld start/restart/stop
```

- ຄໍາສັ່ງຕັ້ງຄ່າການເລີ່ມຕົ້ນMySQL

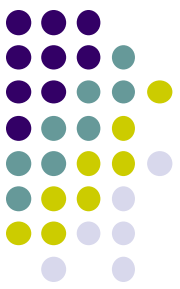
```
# /usr/bin/mysql_secure_installation
```

```
Enter current password for root (enter for none):  
OK, successfully used password, moving on...
```

```
Setting the root password ensures that nobody can log into the MySQL  
root user without the proper authorisation.
```

```
Set root password? [Y/n] █
```

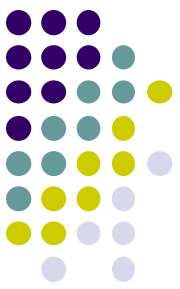
ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL



- ໃສ່ລະຫັດຜ່ານໃຫ້ຜູ້ໃຊ້ root

```
Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!
```

ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL

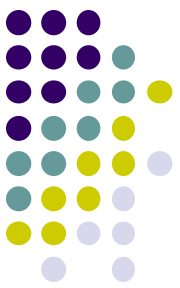


- ລຶບຜູ້ໃຊ້ anonymous

By default, a MySQL installation has an anonymous user, allowing anyone to log into MySQL without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

```
Remove anonymous users? [Y/n] y
... Success!
```

ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL



- ກຳນົດ ໃຫ້ຜູ້ໃຊ້ root ເຂົ້າເຖິງລະບົບຜ່ານເຄືອ

```
Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.
```

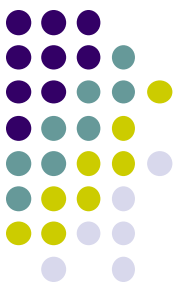
```
Disallow root login remotely? [Y/n] n
... skipping.
```

- ລຶບຖານຂໍ້ມູນຊື່ test

```
Remove test database and access to it? [Y/n] n
... skipping.
```

```
Reloading the privilege tables will ensure that all changes made so far will take effect immediately.
```


ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL



- ກຳນົດສິດທິໃຫ້ກັບຕາຕະລາງ

```
Reloading the privilege tables will ensure that all changes made so far  
will take effect immediately.
```

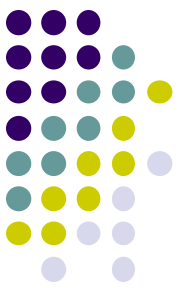
```
Reload privilege tables now? [Y/n] n  
... skipping.
```

```
Cleaning up...
```

```
All done! If you've completed all of the above steps, your MySQL  
installation should now be secure.
```

```
Thanks for using MySQL!
```

ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL



- ການເຂົ້າສູ່ MySQL

```
[root@localhost ~]# mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 5.0.95 Source distribution

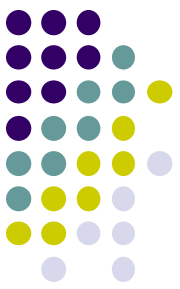
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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ MySQL

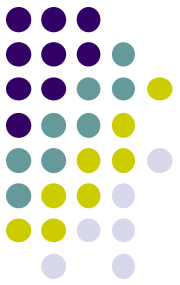


- ຄໍາສັ່ງຕ່າງໆໃນການຈັດການ MySQL

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| test |
+-----+
3 rows in set (0.05 sec)

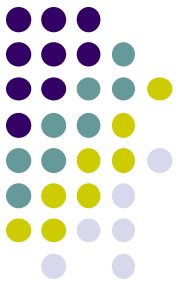
mysql> █
```

ການຈັດການຖານຂໍ້ມູນ



- phpMyAdmin ເປັນເຄື່ອງມືໃນການຈັດການຖານຂໍ້ມູນ MySQL ຜ່ານທາງໜ້າເວບ, ຊຶ່ງງ່າຍຕໍ່ການຈັດການໂດຍນຳໃຊ້ໄດ້ທັງຄຳສັ່ງ ແລະ ຜ່ານໜ້າເວບ, ສາມາດດາວໂຫຼດມານຳໃຊ້ໄດ້ໂດຍບໍ່ຕ້ອງເສຍຄ່າ (Opensource) <http://www.phpmyadmin.net>
- phpMyAdmin ເປັນເວບໂປຣແກຣມທີ່ພັດທະນາດ້ວຍພາສາ PHP ສະນັ້ນກ່ອນຈະນຳໃຊ້ຕ້ອງໄດ້ຕິດຕັ້ງ PHP ແລະ ເວບເຊີເວີ ກ່ອນຈຶ່ງສາມາດຕິດຕັ້ງ phpMyAdmin ໄດ້.

ການຈັດການຖານຂໍ້ມູນ



- phpMyAdmin

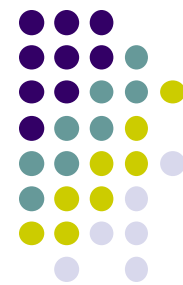
The screenshot shows the phpMyAdmin login page. At the top, there is a logo with a sailboat and the text "phpMyAdmin". Below the logo, it says "Welcome to phpMyAdmin". There are two main sections: a "Language" section with a dropdown menu currently set to "English", and a "Log in" section. The "Log in" section has fields for "Username:" (containing "root") and "Password:". At the bottom right of the login section is a "Go" button.

ການຈັດການຖານຂໍ້ມູນ



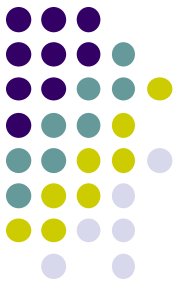
- ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ
 - ດາວໂຫຼດ ແລະ ແຕກໄຟລ `phpmyadmin-4.2.12-all-languages.tar.gz`
`#tar -zxvf phpmyadmin-4.2.12-all-languages.tar.gz`
 - ສຳເນົາ ຫຼື ຍ້າຍໄຟລໄປໄວ້ໃນເວບເຊີເວີ
`#mv phpmyadmin-4.2.12-all-languages`
`/var/www/abc.com/public_html/phpMyAdmin`

ການຈັດການຖານຂໍ້ມູນ



- ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ
 - ສໍາເນົາໄຟລການຕັ້ງຄ່າຕົວຢ່າງ
phpMyAdmin/config.sample.inc.php ແລະ ແກ້ໄຂ
ໄຟລ phpMyAdmin/config.inc.php

```
[root@localhost ~]# cp /var/www/abc.com/public_html/phpMyAdmin/config.sample.inc
.php /var/www/abc.com/public_html/phpMyAdmin/config.inc.php
[root@localhost ~]# vi /var/www/abc.com/public_html/phpMyAdmin/config.inc.php
[root@localhost ~]# █
```



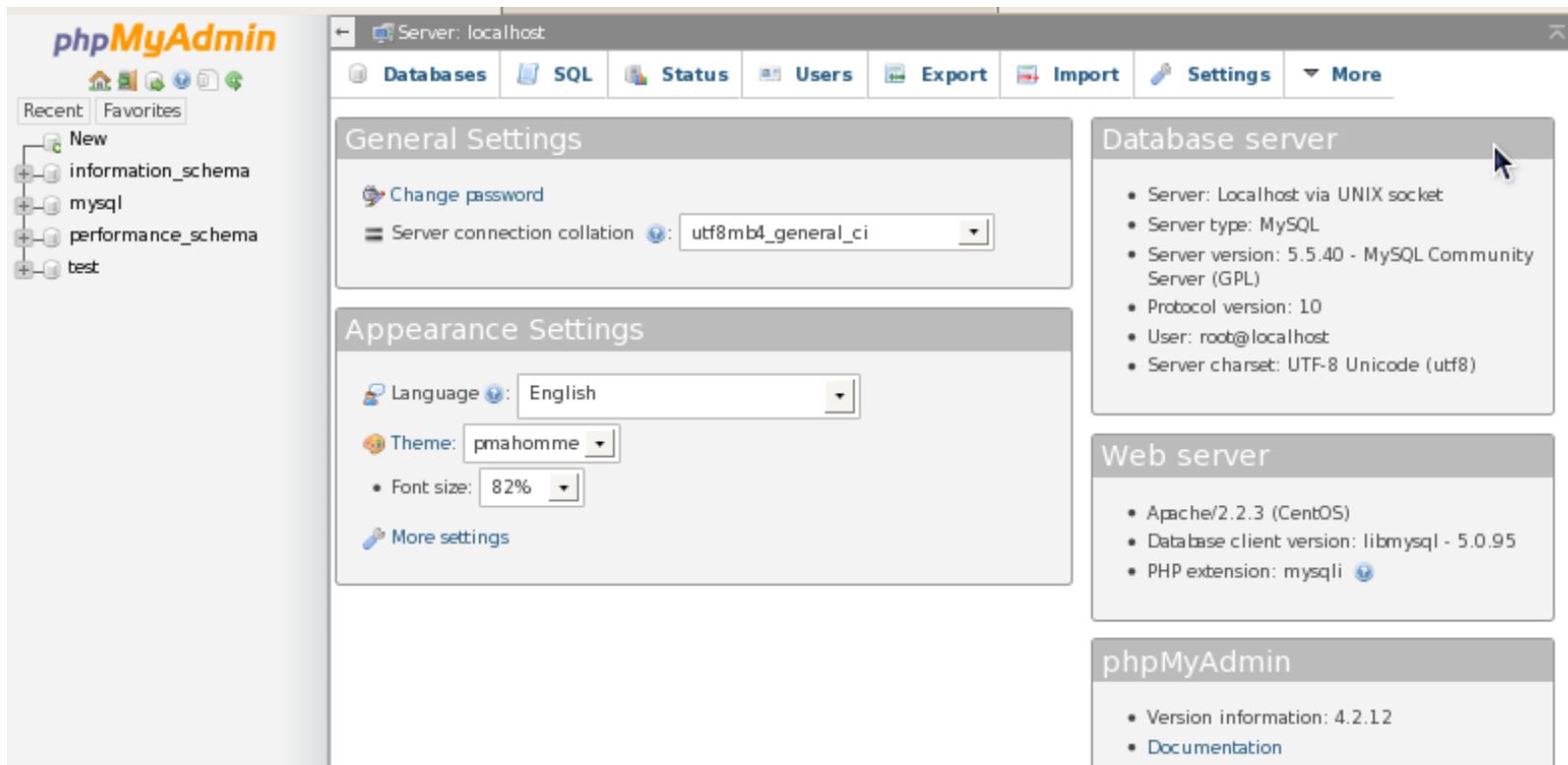
ການຈັດການຖານຂໍ້ມູນ

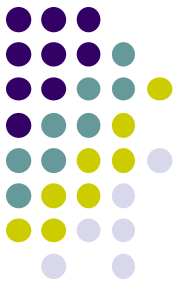
- ການຕິດຕັ້ງ ແລະ ການຕັ້ງຄ່າ(ຕໍ່)

```
* First server
*/
$i++;
/* Authentication type */
$cfg['Servers'][$i]['auth_type'] = 'cookie';
/* Server parameters */
$cfg['Servers'][$i]['host'] = 'localhost';
$cfg['Servers'][$i]['connect_type'] = 'tcp';
$cfg['Servers'][$i]['compress'] = false;
$cfg['Servers'][$i]['AllowNoPassword'] = true;
```


ການຈັດການຖານຂໍ້ມູນ

- ທິດລອງເຂົ້າສູ່ phpMyAdmin



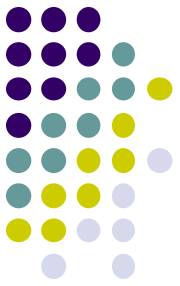


ການນຳໃຊ້ PHP ເຊື່ອມຕໍ່ຖານຂໍ້ມູນ

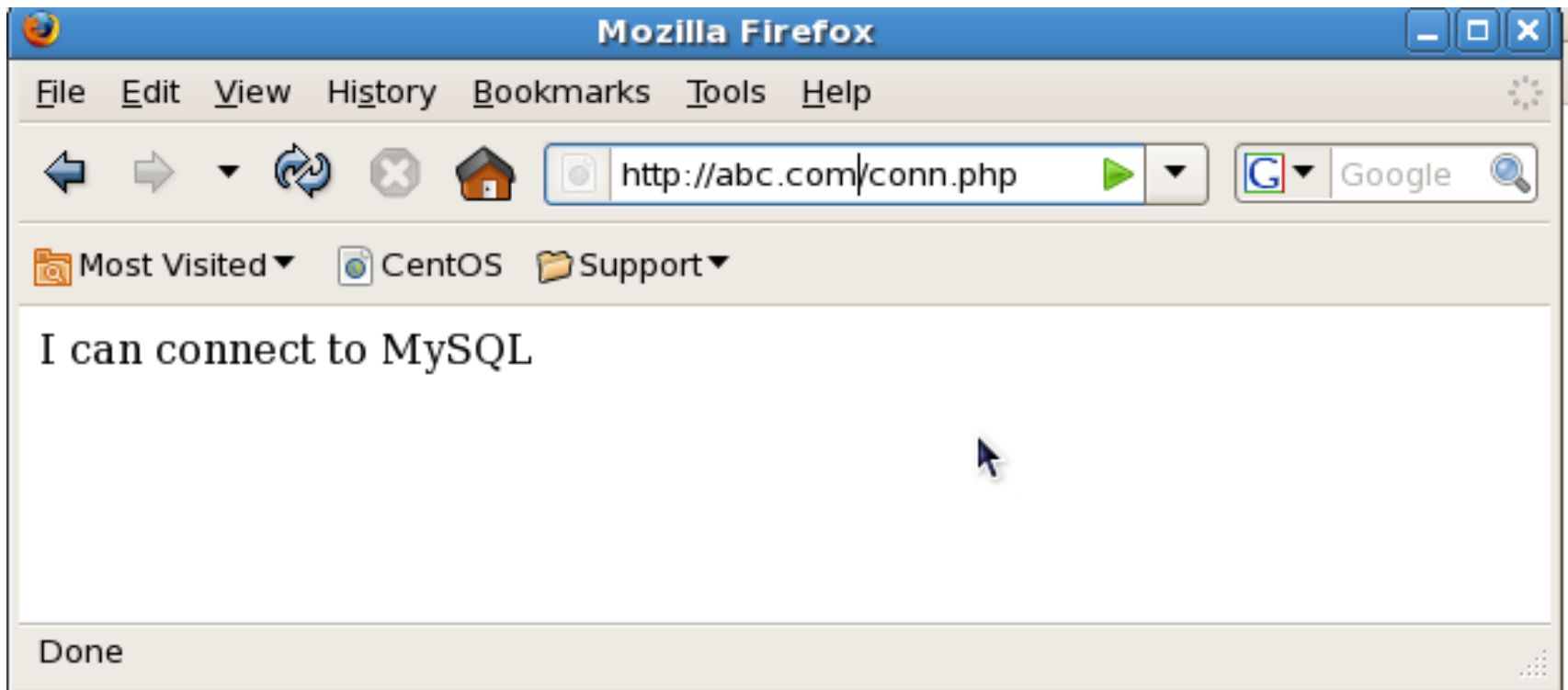
- ສ້າງໄຟລ PHP ເພື່ອທົດສອບຕິດຕໍ່ກັບຖານຂໍ້ມູນ

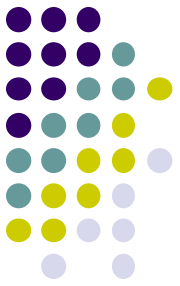
```
File Edit View Terminal Tabs Help
<?php
$conn=mysql_connect("localhost","root","");
mysql_select_db("test");
if($conn){
echo "I can connect to MySQL";
}else{
echo "error";
}
?>
```

ການນຳໃຊ້ PHP ເຊື່ອມຕໍ່ຖານຂໍ້ມູນ



- ທົດສອບການເຊື່ອມຕໍ່ຖານຂໍ້ມູນ





ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows

- ດາວໂຫຼດ MySQL ຈາກ

<https://dev.mysql.com/downloads/installer/>

MySQL Installer 5.6.38

Select Version:

5.6.38

Looking for the latest GA version?

Select Operating System:

Microsoft Windows

Windows (x86, 32-bit), MSI Installer

5.6.38

18.5M

[Download](#)

(mysql-installer-web-community-5.6.38.0.msi)

MD5: 5735b47abfd45e5615f98b982d3aa85d | [Signature](#)

Windows (x86, 32-bit), MSI Installer

5.6.38

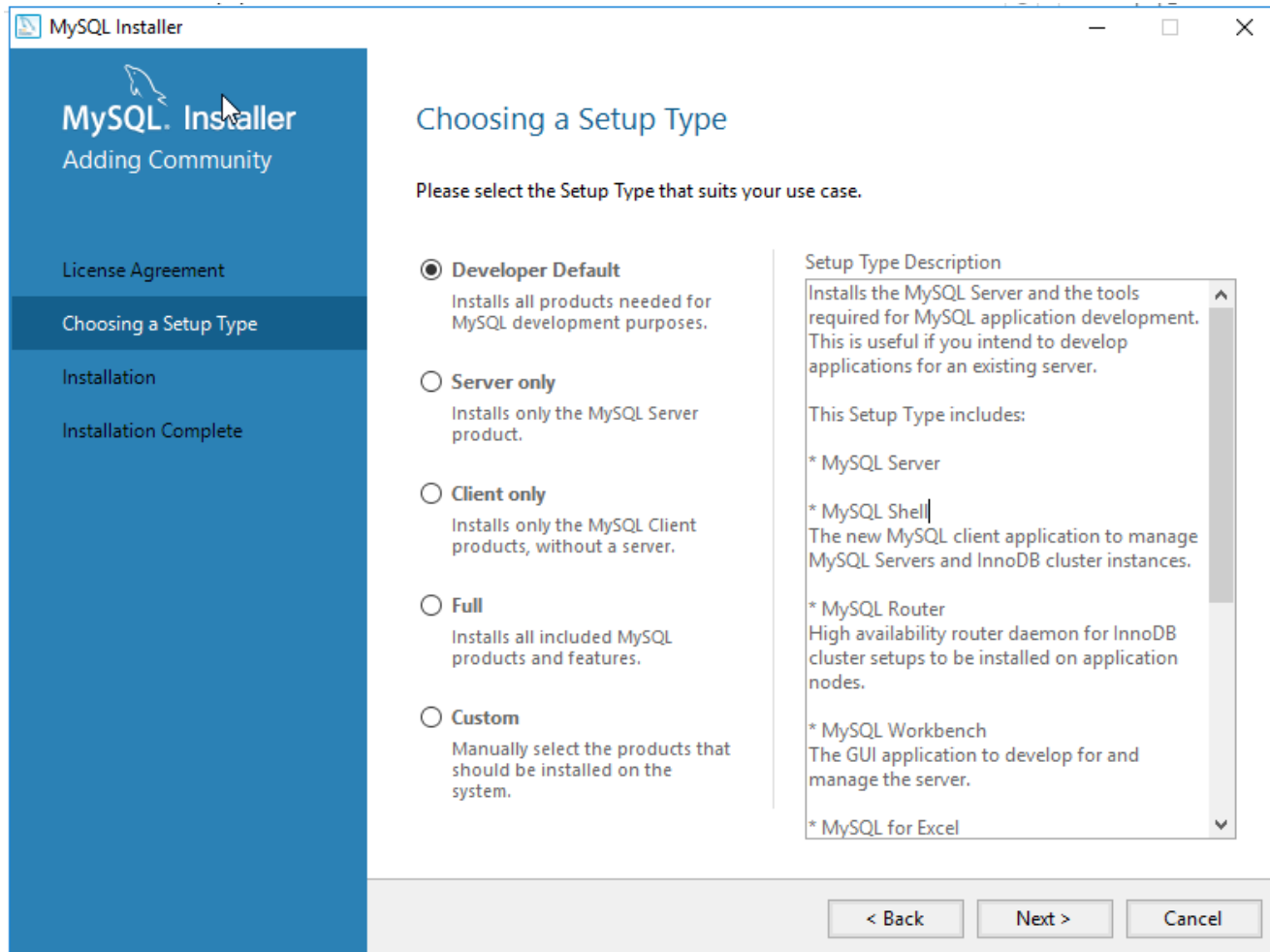
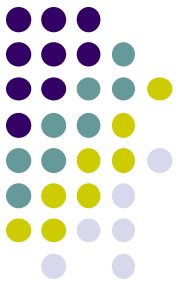
230.8M

[Download](#)

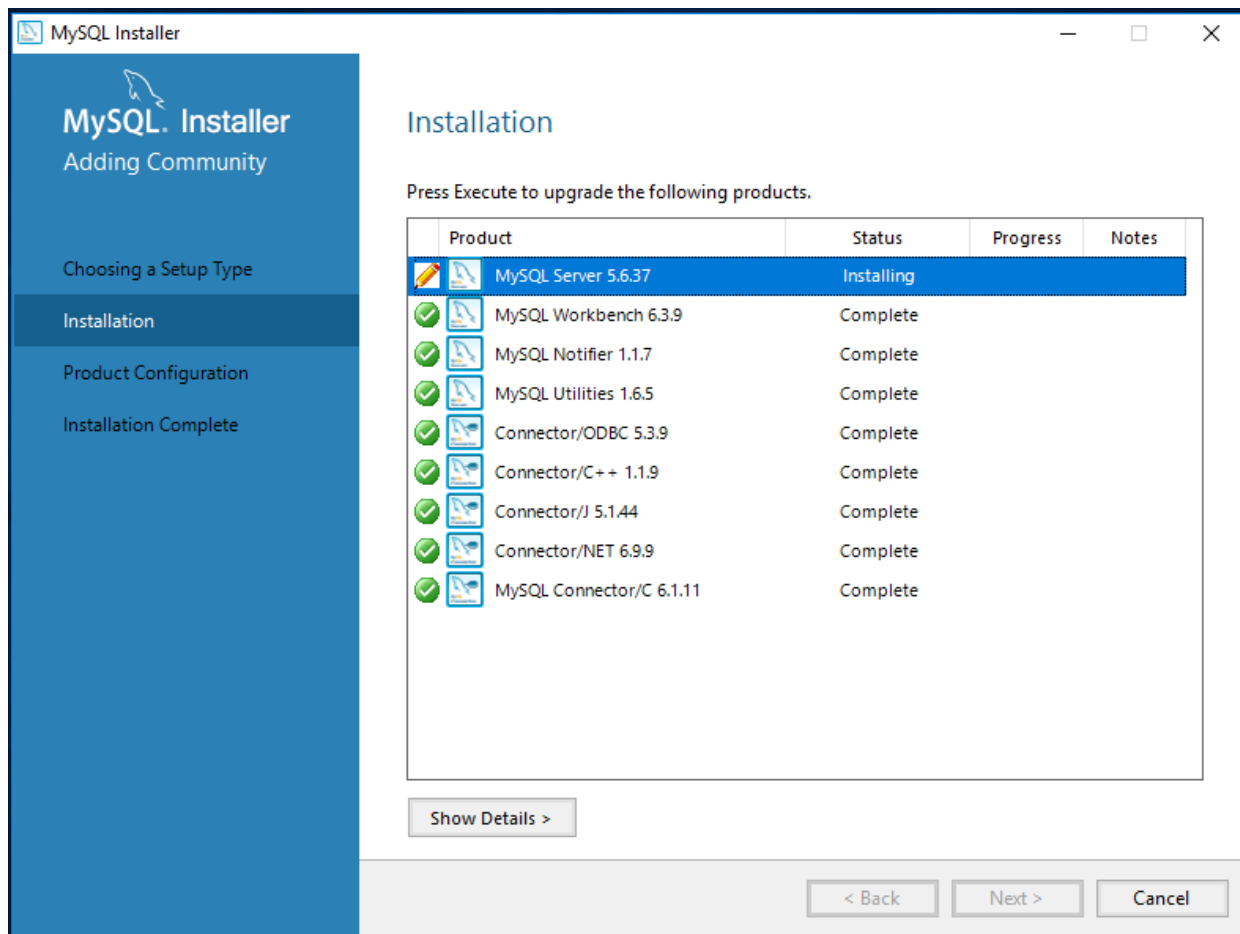
(mysql-installer-community-5.6.38.0.msi)

MD5: caea44e988418708c80106a37f7ea885 | [Signature](#)

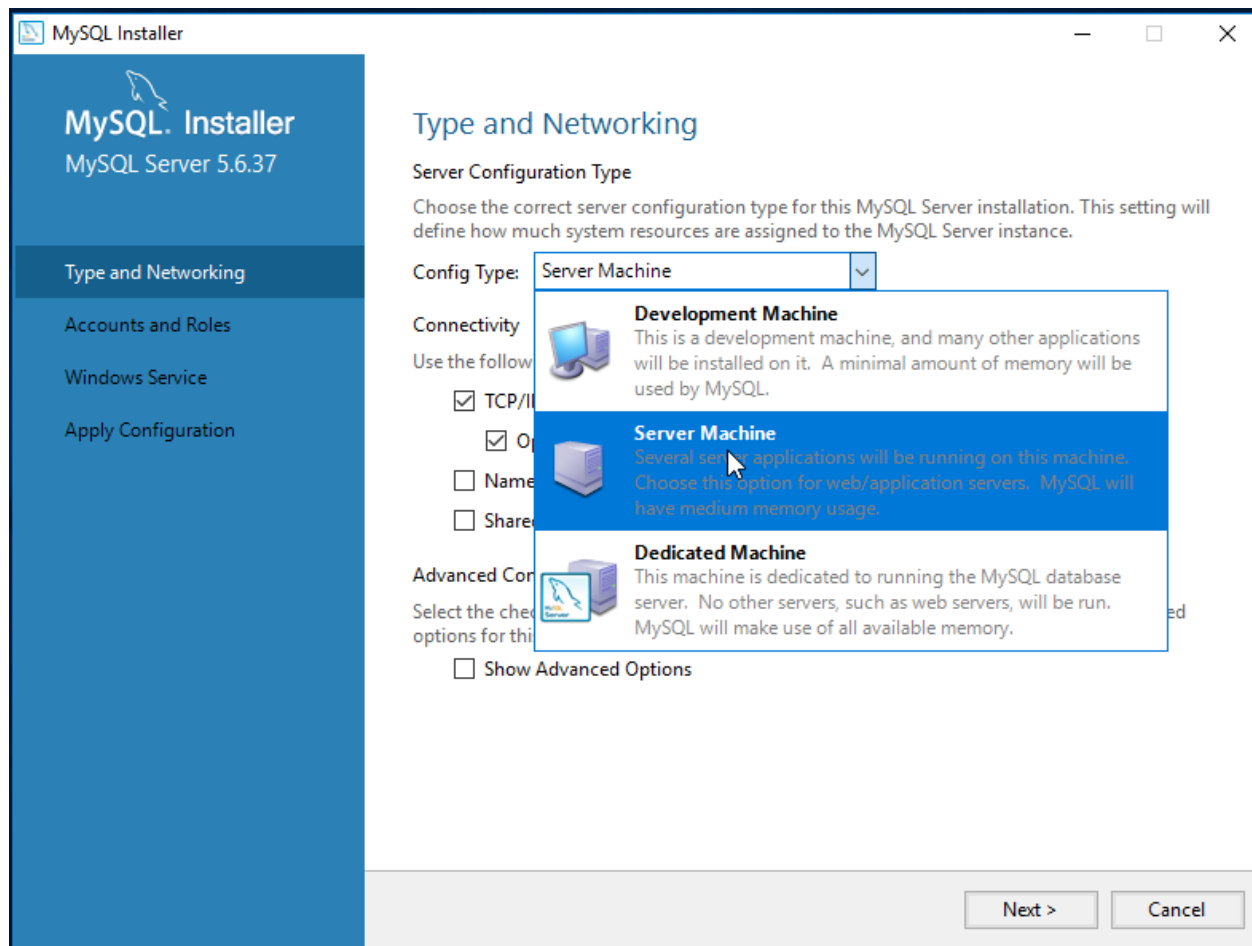
ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows



ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows



ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows



ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows



The screenshot shows the MySQL Installer window for MySQL Server 5.6.37. The left sidebar has four options: 'Type and Networking', 'Accounts and Roles' (which is selected), 'Windows Service', and 'Apply Configuration'. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', asks for a password for the root account. It has two input fields: 'MySQL Root Password' and 'Repeat Password', both containing four dots. Below these fields, the 'Password Strength' is indicated as 'Weak' in red text. The second section, 'MySQL User Accounts', instructs the user to create MySQL user accounts. It features a table with three columns: 'MySQL Username', 'Host', and 'User Role'. To the right of the table are three buttons: 'Add User', 'Edit User', and 'Delete'. At the bottom of the window, there are three buttons: '< Back', 'Next >', and 'Cancel'.

MySQL Installer
MySQL Server 5.6.37

Type and Networking
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password Strength: **Weak**

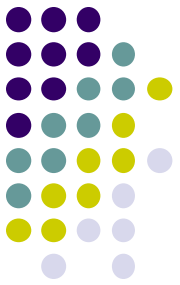
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL Username	Host	User Role
----------------	------	-----------

Add User
Edit User
Delete

< Back Next > Cancel

ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows

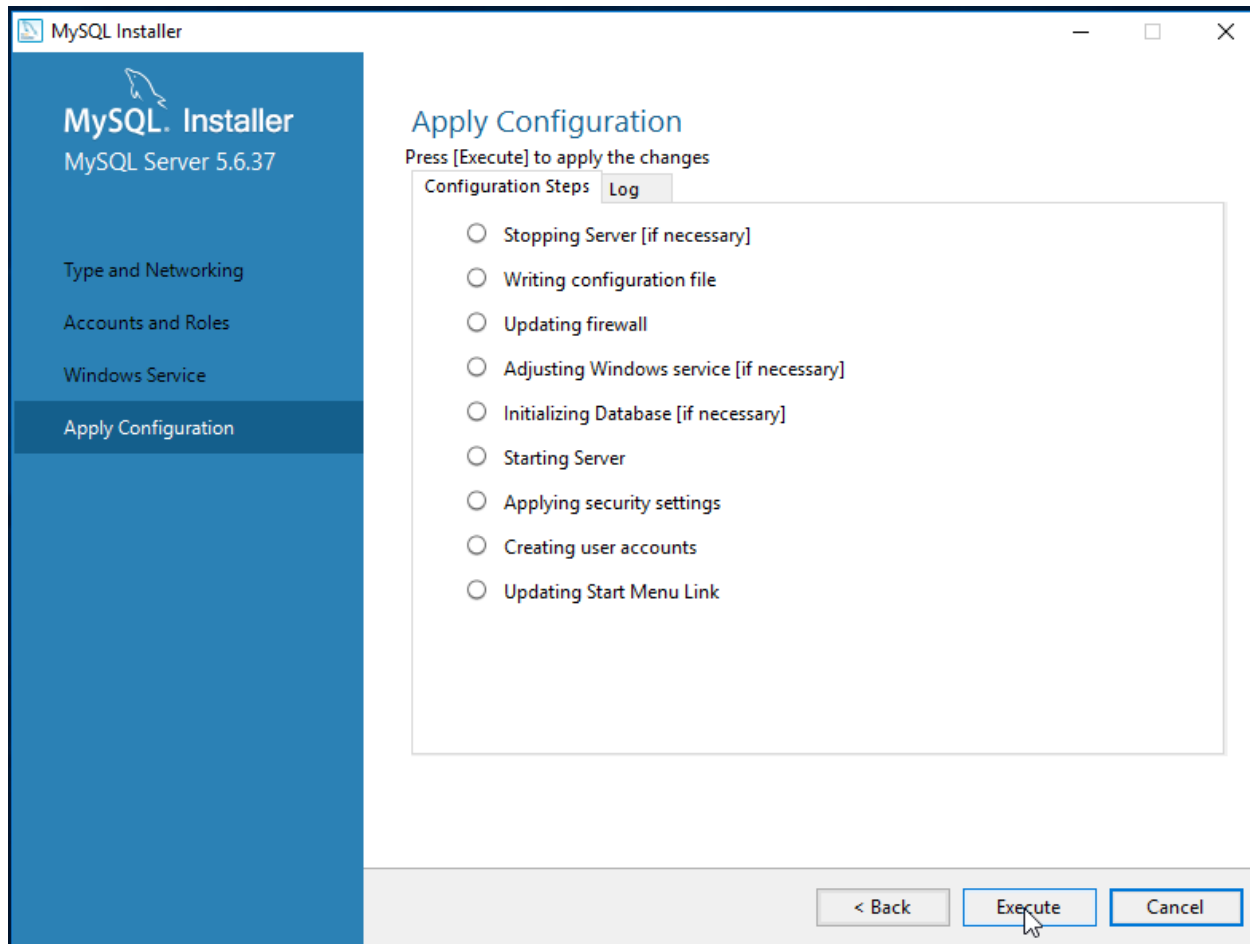
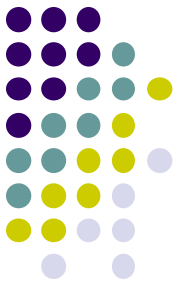


The screenshot shows the 'MySQL Installer' window for 'MySQL Server 5.6.37'. The left sidebar contains a list of configuration steps: 'Type and Networking', 'Accounts and Roles', 'Windows Service' (which is currently selected and highlighted), and 'Apply Configuration'. The main area is titled 'Windows Service' and contains the following options:

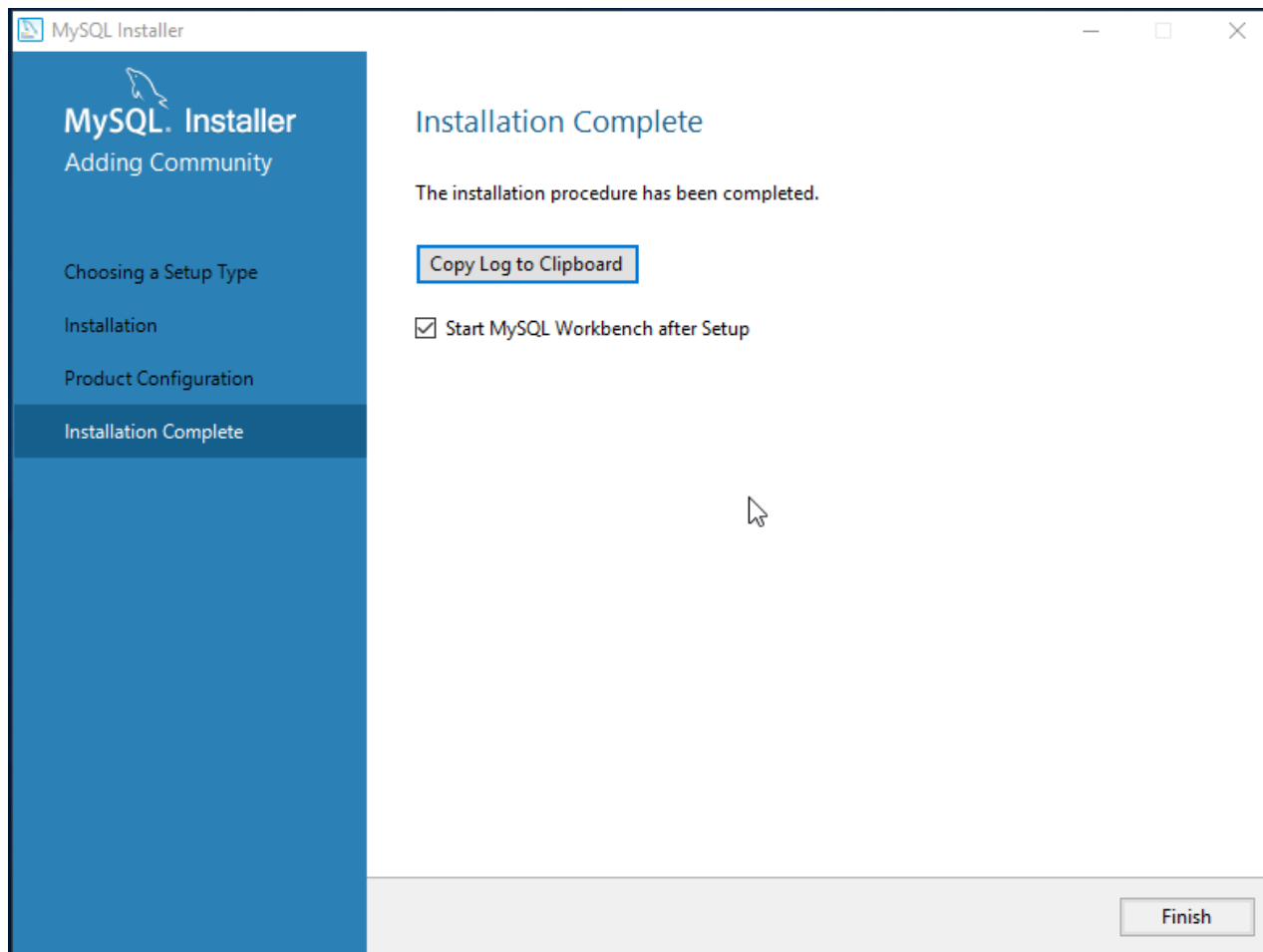
- ☒ **Configure MySQL Server as a Windows Service**
- Windows Service Details**
Please specify a Windows Service name to be used for this MySQL Server instance. A unique name is required for each instance.
Windows Service Name:
- ☒ **Start the MySQL Server at System Startup**
- Run Windows Service as ...**
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.
 - ☒ **Standard System Account**
Recommended for most scenarios.
 - ☐ **Custom User**
An existing user account can be selected for advanced scenarios.

At the bottom right, there are three buttons: '< Back', 'Next >' (which is highlighted with a blue border), and 'Cancel'.

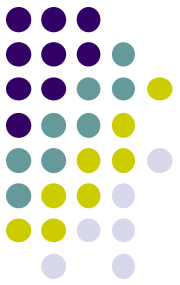
ການນຳໃຊ້ MySQL ຢູ່ໃນ Windows



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```
MySQL 5.6 Command Line Client - Unicode
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 4
Server version: 5.6.37-log MySQL Community Server (GPL)

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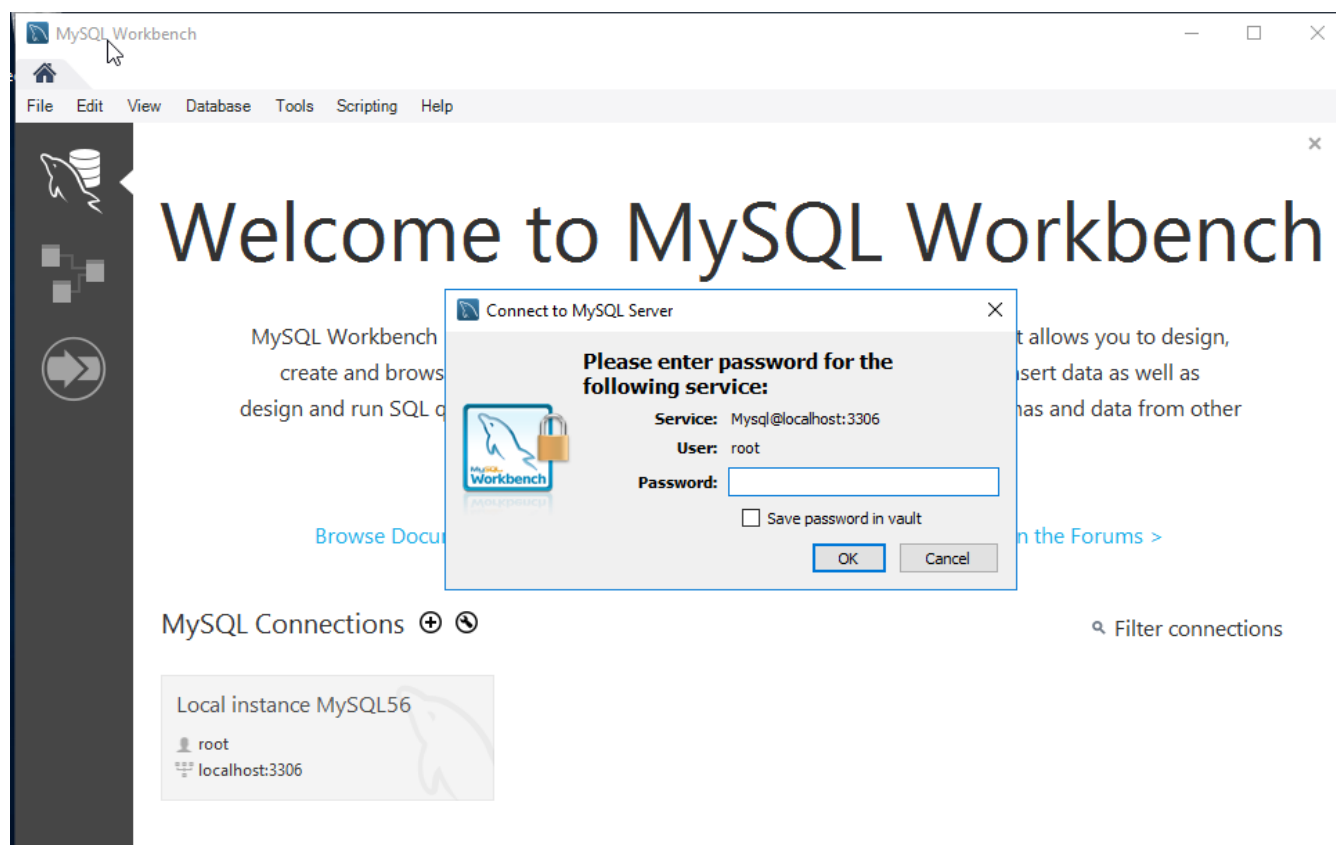
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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement
.

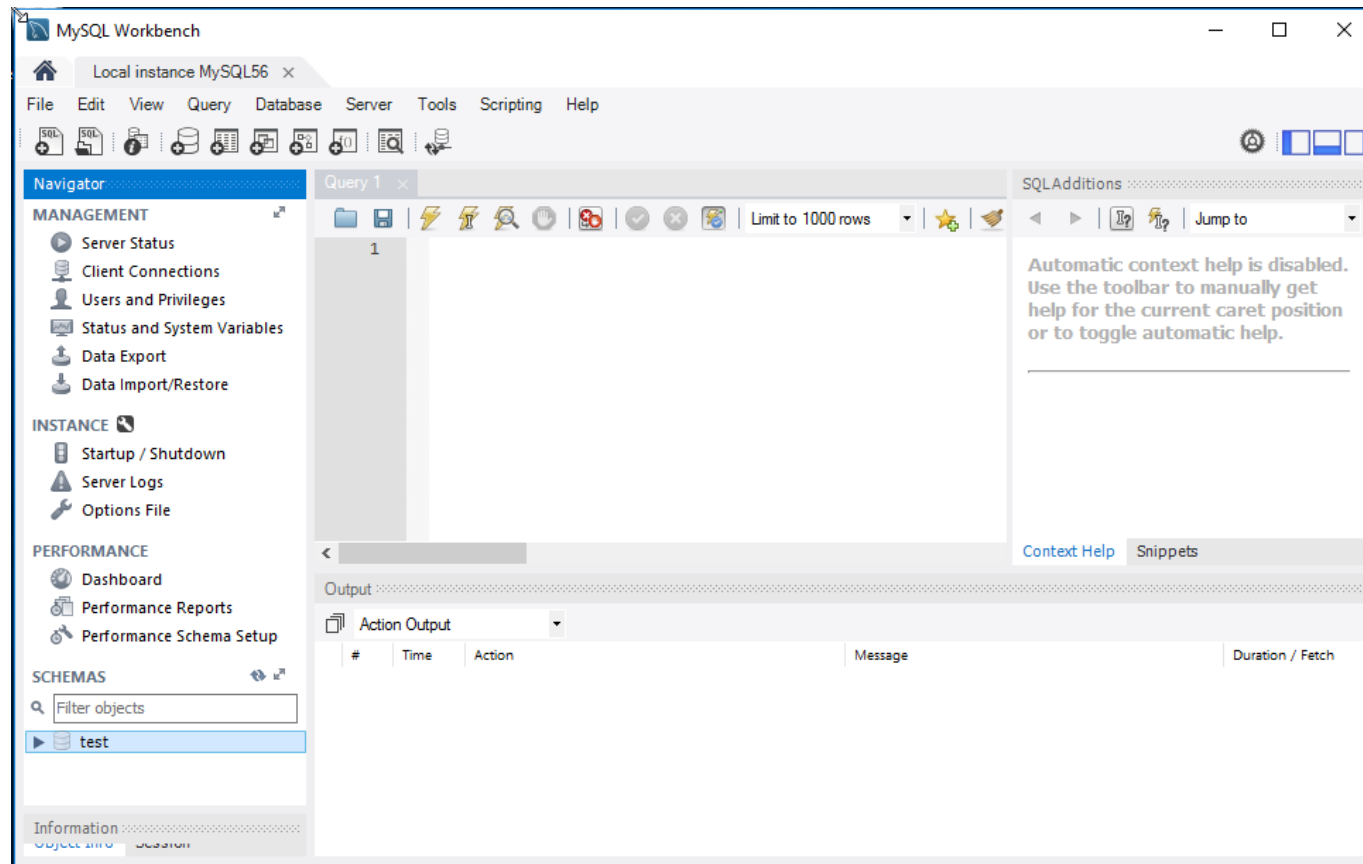
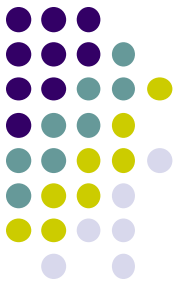
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| test |
+-----+
4 rows in set (0.01 sec)

mysql> _
```

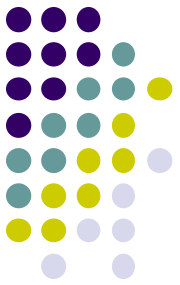
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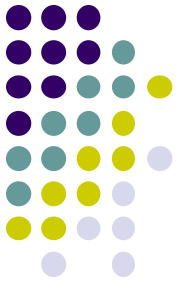
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ຂໍ້ມູນອ້າງອີງ



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- [3] <https://dev.mysql.com/downloads/windows/installer/5.6.html>
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ព្យាបាល និង ព្យាបាល

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